Simulation control

SWAT+ Documentation

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Purpose

`file.cio` is the master index of a SWAT+ project. It is the first file SWAT+ opens, and it must be in the run folder under that name. Each line names the files for one input category, so every other input file is reached through `file.cio`. The last six lines give optional folders for the weather data files and for the output files.

Format

Line 1 is a title and is skipped. Lines 2 to 26 each start with a category label and continue with file names in a fixed order. Lines 27 to 31 each give one weather data folder, and line 32 gives the output folder.

The reader takes the lines by position. The category label is read and discarded, so its text is not checked, and the file names are matched to slots by their order on the line. Values are separated by spaces.

- Write `null` for a file the project does not use. Most readers treat a name that does not exist in the run folder the same way as `null` and skip the file. The exceptions are listed under Missing and null files.
- File names can be up to 25 characters long. Longer names are cut to 25 characters.
- Each line must give at least as many names as its category has slots. If a line is short, the reader continues onto the next line to fill the missing slots, and every later line is then read one line out of step. Extra names at the end of a line are ignored.

Categories and default file names

The labels below are the ones the SWAT+ Editor writes. The default names are used only for slots that are never read, which happens when `file.cio` ends early. A project normally gives every name explicitly.

Line	Label	Files, in slot order (default names)	Documented in
2	simulation	time.sim, print.prt, object.prt, object.cnt, constituents.cs	time.sim, print.prt, object.prt, object.cnt, constituents.cs
3	basin	codes.bsn, parameters.bsn, carbon.bsn	codes.bsn, parameters.bsn, carbon.bsn
4	climate	weather-sta.cli, weather-wgn.cli, pet.cli, pcp.cli, tmp.cli, slr.cli, hmd.cli, wnd.cli, atmodep.cli	climate index files
5	connect	hru.con, hru-lte.con, rout_unit.con, gwflow.con, aquifer.con, aquifer2d.con, channel.con, reservoir.con, recall.con, exco.con, delratio.con, outlet.con, chandeg.con	connectivity files
6	channel	initial.cha, channel.cha, hydrology.cha, sediment.cha, nutrients.cha, channel-lte.cha, hyd-sed-lte.cha, temperature.cha	legacy channel files, channel tables
7	reservoir	initial.res, reservoir.res, hydrology.res, sediment.res, nutrients.res, weir.res, wetland.wet, hydrology.wet	reservoir files, wetland files
8	routing_unit	rout_unit.def, rout_unit.ele, rout_unit.rtu, rout_unit.dr	connectivity files
9	hru	hru-data.hru, hru-lte.hru	hru-data.hru
10	exco	exco.exc, exco_om.exc, exco_pest.exc, exco_path.exc, exco_hmet.exc, exco_salt.exc	connectivity files
11	recall	recall.rec	recall
12	dr	delratio.del, dr_om.del, dr_pest.del, dr_path.del, dr_hmet.del, dr_salt.del	connectivity files
13	aquifer	initial.aqu, aquifer.aqu	aquifer files
14	herd	animal.hrd, herd.hrd, ranch.hrd	not read
15	water_rights	place_of_use.wro, point_of_diver.wro	water allocation
16	link	chan-surf.lin, aqu_cha.lin	Link files
17	hydrology	hydrology.hyd, topography.hyd, field.fld	hydrology and topography
18	structural	tiledrain.str, septic.str, filterstrip.str, grassedww.str, bmpuser.str	structural BMPs
19	hru_parm_db	plants.plt, fertilizer.frt, tillage.til, pesticide.pes, pathogens.pth, metals.mtl, salt.slt, urban.urb, septic.sep, snow.sno	plants.plt, fertilizer.frt, tillage.til, pesticide.pes, pathogens.pth, metals.mtl, salt inputs, urban.urb and septic.sep, snow.sno
20	ops	harv.ops, graze.ops, irr.ops, chem_app.ops, fire.ops, sweep.ops	operation databases
21	lum	landuse.lum, management.sch, cntable.lum, cons_practice.lum, ovn_table.lum	landuse.lum, management.sch, lookup tables
22	chg	cal_parms.cal, calibration.cal, codes.sft, wb_parms.sft, water_balance.sft, ch_sed_budget.sft, ch_sed_parms.sft, plant_parms.sft, plant_gro.sft	parameter change files, soft calibration
23	init	plant.ini, soil_plant.ini, om_water.ini, pest_hru.ini, pest_water.ini, path_hru.ini, path_water.ini, hmet_hru.ini, hmet_water.ini, salt_hru.ini, salt_water.ini	initial conditions
24	soils	soils.sol, nutrients.sol, soils_lte.sol	soils.sol
25	decision_table	lum.dtl, res_rel.dtl, scen_lu.dtl, flo_con.dtl	decision tables
26	regions	ls_unit.ele, ls_unit.def, ls_reg.ele, ls_reg.def, ls_cal.reg, ch_catunit.ele, ch_catunit.def, ch_reg.def, aqu_catunit.ele, aqu_catunit.def, aqu_reg.def, res_catunit.ele, res_catunit.def, res_reg.def, rec_catunit.ele, rec_catunit.def, rec_reg.def	aquifer files, recall, soft calibration
27	pcp_path	folder for the precipitation data files	precipitation data

Line	Label	Files, in slot order (default names)	Documented in
28	`tmp_path`	folder for the temperature data files	temperature data
29	`slr_path`	folder for the solar radiation data files	climate index files
30	`hmd_path`	folder for the relative humidity data files	climate index files
31	`wnd_path`	folder for the wind speed data files	climate index files
32	any word	output folder	Output folder

Unused slots

These names are read from `file.cio`, but SWAT+ 62.0.1 never opens the file they name. Any name or `null` can be given.

- `connect`: `aquifer2d.con` (slot 6).
- `routing_unit`: `rout_unit.dr` (slot 4).
- `herd`: all three slots.
- `hru_parm_db`: `metals.mtl` and `salt.slt` (slots 6 and 7).
- `init`: `hmet_water.ini`, `salt_hru.ini` and `salt_water.ini` (slots 9 to 11).
- `regions`: `ls_cal.reg` and `ch_catunit.ele` (slots 5 and 6).

Two of these files are read under a fixed name instead. When salts are simulated, the HRU salt initial conditions come from a file named `salt_hru.ini` in the run folder, whatever slot 10 of the `init` line says. Channel catchment unit elements come from a file named `element.ccu`.

The `climate` line has no wind direction file. It has nine slots, and slot 3 is `pet.cli`.

Missing and null files

For most slots, a name of `null` or a name that does not exist in the run folder means the file is skipped and the related feature is off. Some files behave differently:

- `object.cnt`: if it is missing or `null`, SWAT+ prints `Cannot find object.cnt input file` and stops.
- `carbon.bsn`: read only when `carbon = 2` in `codes.bsn`. It is then required, and SWAT+ stops with an error, written to the screen and to `diagnostics.out`, if the file named in slot 3 of the `basin` line does not exist or cannot be opened. The per-layer carbon file has no slot on the line: its name is built from the carbon file name, so `carbon.bsn` gives `carbon_lyr.bsn`.
- `time.sim` and `print.prt`: there is no check for a missing file. If the name is not `null` and the file does not exist, SWAT+ creates an empty file of that name and continues with built-in values. For `time.sim` this leaves the run without a valid simulation period, so the file is required in practice.
- `place_of_use.wro` and `point_of_diver.wro`: if either is missing or `null`, SWAT+ creates an empty file of that name (a file called `null` in the second case), reads nothing from it, and water allocation is off. See water allocation.
- `chan-surf.lin`: if the name is not `null` and the file does not exist, SWAT+ creates an empty file of that name. See link files.

Weather data folders

Lines 27 to 31 let the precipitation, temperature, solar radiation, humidity and wind data files be kept outside the run folder. Each line has a label and one value of up to 80 characters. A value of `null` means the data files are in the run folder.

SWAT+ joins the folder and the file name from the climate index file (for example `pcp.cli`) without adding a separator, so the folder must end with `/` or `\`. The folder is read in the same way as a file name, with two consequences:

- A folder name that contains spaces must be in quotes.
- An unquoted `/` ends the value. `data/pcp/` is read as `data`, and an unquoted value that starts with `/` is read as empty, so the files are looked for in the run folder. Put any folder that uses forward slashes in quotes, for example `"C:/swat/weather/pcp/"`, or use backslashes on Windows.

There is no folder line for potential evapotranspiration data. Files listed in pet.cli are always read from the run folder.

Output folder

Line 32 is optional. It has a label (any word) and a folder name. SWAT+ reads the whole line, drops the first word and takes the rest of the line as the folder, so the folder name can contain spaces and forward slashes without quotes. Up to 256 characters are kept. If the line is missing or empty, or the folder is `null` or `NULL`, the output files are written to the run folder.

When a folder is given, SWAT+ converts the separators to the ones the operating system uses, creates the folder and any missing parent folders, and prints `Output path set to:` with the folder name. A relative folder is taken from the run folder. On Linux and macOS a folder that contains `:` is treated as a Windows path, and SWAT+ stops with an error. On Windows a folder that starts with `/` is placed on drive `C:`. SWAT+ also stops if it cannot create or open the folder.

The standard output files go to this folder, including `files_out.out`, `diagnostics.out`, `checker.out` and the time series outputs switched on in print.prt. In 62.0.1 the gwflow output files and the soft calibration output files (`hru-out.cal`, `hru-new.cal` and others) are still written to the run folder.

Older file.cio files

Two lines changed length in recent revisions, and a `file.cio` written for an earlier revision needs both updated before it is run with 62.0.1:

- The `basin` line now has three slots. Add carbon.bsn or `null` after parameters.bsn. With only two names on the `basin` line, SWAT+ takes the first word of line 4 (`climate`) as the carbon file name and discards the rest of that line. Every later category is then read from the line below the one it belongs to: the climate files are taken from the `connect` line, the connection files from the `channel` line, and so on. No message reports the shift. If `carbon = 2` in codes.bsn, SWAT+ stops with `ERROR: climate is required when carbon is enabled`; otherwise the run continues with wrong file names in every later slot.
- The `water_rights` line (line 15) now has two slots, the place-of-use file and the point-of-diversion file. An older line with `water_allocation.wro element.wro water_rights.wro` makes SWAT+ read `water_allocation.wro` as the place-of-use file and `element.wro` as the point-of-diversion file, and the third name is ignored. See water allocation for those files.

Example

`refdata/Osu_1hru/file.cio` in the SWAT+ repository (long lines cut at the right):

```
file.cio: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
simulation        time.sim          print.prt         null              object.cnt        null
basin             codes.bsn         parameters.bsn    null
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli           tmp.cli
connect           hru.con           null              rout_unit.con     null              aquifer.con
channel           initial.cha       null              null              null              nutrients.cha
reservoir         initial.res       null              null              sediment.res      nutrients.res
routing_unit      rout_unit.def     rout_unit.ele     rout_unit.rtu     null
hru               hru-data.hru      null
exco              null              null              null              null              null
recall            recall.rec
dr                null              null              null              null              null
aquifer           initial.aqu       aquifer.aqu
herd              null              null              null
water_rights      null              null              null
link              null              null
hydrology         hydrology.hyd     topography.hyd    field.fld
```

structural	tiledrain.str	septic.str	filterstrip.str	grassedww.str	bmpuser.str
hru_parm_db	plants.plt	fertilizer.frt	tillage.til	pesticide.pes	null
ops	harv.ops	graze.ops	irr.ops	chem_app.ops	fire.ops
lum	landuse.lum	management.sch	cntable.lum	cons_practice.lum	ovn_table.lum
chg	cal_parms.cal	calibration.cal	null	null	null
init	plant.ini	soil_plant.ini	om_water.ini	null	null
soils	soils.sol	nutrients.sol	null		
decision_table	lum.dtl	res_rel.dtl	null	null	
regions	ls_unit.ele	ls_unit.def	null	null	null
pcp_path	null				
tmp_path	null				
slr_path	null				
hmd_path	null				
wnd_path	null				

The `basin` line has three slots, with `null` for the carbon file because the project does not use the carbon model. The `climate` line uses measured precipitation, temperature, solar radiation and wind; the PET, humidity and atmospheric deposition files are not used. The `water_rights` line still has the older three slots, all `null`; the first two switch water allocation off and the third is ignored. All weather data files are in the run folder, and there is no output folder line, so the outputs are written to the run folder.

In `refdata/Ames_sub1/file.cio` the `aquifer` line reads `aquifer nbull null`. `nbull` is taken as a file name. No file of that name exists, so the initial aquifer file is skipped in the same way as for `null`.

Related files

- `time.sim`, `print.prt` and `object.cnt`: the simulation control files named on line 2.
- `codes.bsn`: its `carbon` code decides whether the carbon file on line 3 is read.
- Input reference: every input file reached from `file.cio`.

Sources

- Reader: `readcio_read.f90`, called first from `proc_bsn.f90`
- Slot names and defaults: `input_file_module.f90`
- Output folder: `output_path_module.f90`
- Weather folders: `cli_pmeas.f90`, `cli_tmeas.f90`, `cli_smeas.f90`, `cli_hmeas.f90`, `cli_wmeas.f90`, `cli_petmeas.f90`
- Carbon file check: `carbon_bsn_read.f90`
- Water allocation files: `water_allocation_read.f90`
- Fixed-name readers: `ch_read_elements.f90` (`element.ccu`), `salt_hru_read.f90` (`salt_hru.ini`)

Line	Fortran variable
2 `simulation`	`in_sim` (`time`, `prt`, `object_prt`, `object_cnt`, `cs_db`)
3 `basin`	`in_basin` (`codes_bas`, `parms_bas`, `carbon_bsn`)
4 `climate`	`in_cli`
5 `connect`	`in_con`
6 `channel`	`in_cha`
7 `reservoir`	`in_res`
8 `routing_unit`	`in_ru`
9 `hru`	`in_hru`
10 `exco`	`in_exco`
11 `recall`	`in_rec`
12 `dr`	`in_delr`
13 `aquifer`	`in_aqu`
14 `herd`	`in_herd`
15 `water_rights`	`in_wallo` (`pou`, `pod`)
16 `link`	`in_link`
17 `hydrology`	`in_hyd`
18 `structural`	`in_str`

Line	Fortran variable
19 `hru_parm_db`	`in_parmdb`
20 `ops`	`in_ops`
21 `lum`	`in_lum`
22 `chg`	`in_chg`
23 `init`	`in_init`
24 `soils`	`in_sol`
25 `decision_table`	`in_cond`
26 `regions`	`in_regs`
27 to 31	`in_path_pcp%pcp`, `in_path_tmp%tmp`, `in_path_slr%slr`, `in_path_hmd%hmd`, `in_path_wnd%wnd`
32	`out_path` in `output_path_module`

Purpose

`time.sim` sets the first and last day of the simulation and the number of time steps per day. Weather input, routing and every output file follow this period. The file is named in slot 1 of the `simulation` line of `file.cio` (default `time.sim`).

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Line 3 contains five integers in the order below. The reader does not match columns by header name.

#	Column	Units	Default	Description
1	`day_start`	day of year	1	Day of the year on which the simulation starts, 1 to 366. A value of 0 or less is set to 1 (1 January).
2	`yrc_start`	year	-	Calendar year in which the simulation starts.
3	`day_end`	day of year	0	Day of the year on which the simulation ends in `yrc_end`. A value of 0 or less means the last day of that year (31 December).
4	`yrc_end`	year	-	Calendar year in which the simulation ends.
5	`step`	time steps per day	1	Number of time steps per day for precipitation, surface runoff and routing. A value of 0 or less is set to 1 (daily).

Simulation period

The simulation runs over the calendar years `yrc_start` to `yrc_end`, which is

$$N_{yr} = \text{yrc_end} - \text{yrc_start} + 1$$

years. The first year starts on `day_start`, and every later year starts on 1 January. Every year but the last runs to 31 December. The last year ends on `day_end`, or on 31 December if `day_end` is 0. A `day_end` of 366 in a year that is not a leap year is treated as day 365. Leap years follow the Gregorian rule: years divisible by 4 are leap years, except century years that are not divisible by 400.

`day_start` and `day_end` make it possible to simulate water years or other periods that do not follow the calendar year. For example, `day_start = 274`, `yrc_start = 1999`, `day_end = 273` and `yrc_end = 2019` runs from 1 October 1999 to 30 September 2019. Neither year is a leap year, so day 274 is 1 October and day 273 is 30 September; in a leap year the same dates are days 275 and 274.

The warm-up years and the print window for daily output are set in `print.prt`. A zero `yrc_start` or `yrc_end` in `print.prt` is filled from this file.

Time step

`step` is the number of time steps in a day:

`step`	Time step
0 or 1	1 day
2	12 hours
24	1 hour
96	15 minutes
1440	1 minute

Any positive whole number works. The length of one time step in minutes is

$$\Delta t = \frac{1440}{\text{step}}$$

With `step` greater than 1, the day is split into sub-daily steps for precipitation, infiltration and surface runoff, hydrograph routing through channels and reservoirs, and sub-daily channel output. The Green and Ampt infiltration method (`gampt = 1` in `codes.bsn`) uses the sub-daily rainfall. Other processes, such as plant growth, soil water and nutrient cycling, still run once a day.

Sub-daily precipitation records (`*.pcp` files with a time step above 0) must have the same number of values per day as `step`. SWAT+ stores `step` values per day and does not check the record against it.

When `step` is greater than 1 and precipitation is generated, the weather generator also distributes each day's rainfall over the time steps. For a station with measured daily records, 62.0.1 does not fill the sub-daily values: the daily total is read, but the rainfall in each time step stays at zero. A sub-daily run therefore needs sub-daily precipitation records or generated precipitation.

Weather outside the observed record

When the simulation starts before, or ends after, the measured daily weather records listed in the climate index files, SWAT+ generates the weather for the days outside the record with the weather generator (`weather-wgn.cli`). The same applies to days that are flagged as missing in a record.

Example

`refdata/Ames_sub1/time.sim` in the SWAT+ repository:

```
time.sim:
day_start  yrc_start   day_end   yrc_end      step
        0       1975         0      2020         0
```

The simulation runs from 1 January 1975 to 31 December 2020, 46 years. `day_start` and `day_end` are 0, so the first and last days of the year are used. `step` is 0, which SWAT+ sets to 1, so the simulation is daily.

Related files

- `file.cio`: names this file.
- `print.prt`: warm-up years and print window for the outputs inside this period.
- `pcp.cli` and the precipitation data files: their time step must match `step` for sub-daily records.
- `codes.bsn`: selects Green and Ampt infiltration, which uses sub-daily rainfall.

Sources

- Reader: `time_read.f90`
- Type definition: `type time_current` in `time_module.f90`

- Time step length: proc_bsn.f90
- Year and day loop, leap years, end day: time_control.f90
- Measured and generated precipitation: cli_pmeas.f90, cli_precip_control.f90
- File name slot: `in_sim%time` in input_file_module.f90

Column	Fortran variable
day_start	time%day_start
yrc_start	time%yrc_start
day_end	time%day_end
yrc_end	time%yrc_end
step	time%step

Purpose

`print.prt` controls the output of a SWAT+ run. It sets the number of warm-up years left out of the outputs, the period and spacing of daily output, the extra output formats, a few special output files, and one row per output group with switches for daily, monthly, yearly and average annual files. The file is named in slot 2 of the `simulation` line of file.cio (default `print.prt`).

If the name in file.cio is `null`, SWAT+ uses the built-in values: no warm-up, every output group off, and crop yields written as average annual values only. A name that does not exist in the run folder has the same effect, and SWAT+ creates an empty file of that name. The files listed in Output naming convention are written in every case.

Format

Free format, whitespace-separated. The file has four short blocks, each a header line followed by a line of values, then a header line and one row per output group. Line 1 is a title. All header lines are skipped, and values are read by position within each line.

Line	Content
1	title
2	header
3	`nyskip`, `day_start`, `yrc_start`, `day_end`, `yrc_end`, `interval`
4	header
5	`aa_int_cnt`, followed by `aa_int_cnt` end years
6	header
7	`csvout`, `use_obj_lbls`, `cdfout`
8	header
9	`crop_yld`, `mgtout`, `hydcon`, `fdcout`
10	header (`objects daily monthly yearly avann`)
11 and on	output group rows

Print period (line 3)

#	Column	Units	Default	Description
1	`nyskip`	years	0	Number of years at the start of the simulation left out of all outputs (warm-up period).
2	`day_start`	day of year	1	Day of the year on which daily output starts. 0 means 1.
3	`yrc_start`	year	first simulated year	Year in which daily output starts. 0 means the first year in time.sim.
4	`day_end`	day of year	366	Day of the year on which daily output ends. 0 means 366, the end of the year.

#	Column	Units	Default	Description
5	`yrc_end`	year	after the last simulated year	Year in which daily output ends. 0 means daily output runs to the end of the simulation.
6	`interval`	days	1	Days between daily output rows. 0 or less means 1. With 2, every other day is written.

A warm-up period lets soil water, nutrient pools and plant cover settle from their initial values before results are used. It matters more for short simulations; for a simulation of five years or less, a warm-up period is recommended.

Average annual intervals (line 5)

#	Column	Units	Default	Description
1	`aa_int_cnt`	count	0	Number of average annual print intervals. When it is above 0, the end year of each interval follows on the same line, in order.

For example, `3 1995 2005 2015` would end intervals in 1995, 2005 and 2015. In 62.0.1 the end years are read but have no effect on the outputs: every average annual file covers the whole printed period. See Average annual values.

Output formats (line 7)

#	Column	Values	Default	Description
1	`csvout`	`y`, `n`	`n`	`y` also writes a comma-separated `.csv` copy of most output files.
2	`use_obj_lbls`	`y`, `n`	`n`	How the output group rows are read: `y` by name, `n` by position. See Output group rows.
3	`cdfout`	`y`, `n`	`n`	netCDF output. Read and not used.

`print.prt` files written by older versions of the SWAT+ Editor have the header `csvout dbout cdfout` on line 6. SWAT+ reads the second value as `use_obj_lbls` whatever the header says, so an old file with `n` there is read by position.

Special outputs (line 9)

#	Column	Values	Default	Description
1	`crop_yld`	`a`, `y`, `b`, `n`	`a`	Crop yield files: `a` average annual (`crop_yld_aa`), `y` yearly (`crop_yld_yr` and the basin crop yield files), `b` both, `n` none.
2	`mgtout`	`y`, `n`	`n`	`y` writes `mgt_out.txt`, the log of management operations, decision table actions and water allocation deliveries.
3	`hydcon`	`y`, `n`	`n`	`y` writes `hydcon.out`, the list of hydrograph connections between objects. It is mostly useful for finding connection errors.
4	`fdcout`	`y`, `n`	`n`	`y` computes flow duration curves and a flashiness index for SWAT+ channels and writes them to `flow_duration_curve.out`.

Older Editor versions write the header `soilout` in place of `crop_yld`; the value is read as `crop_yld`.

Output group rows

Each row from line 11 on has a group name and four switches, for the daily, monthly, yearly and average annual files of that group. A switch of `y` opens the file for that interval and any other value leaves it off. The legacy carbon rows `hru_cb` and `hru_cb_vars` also accept `l`. The files each row opens, and the pages that describe them, are listed in Output naming convention.

How the rows are read depends on `use_obj_lbls` on line 7:

- `use_obj_lbls = y`: each row is matched by its name, so rows can be in any order and rows can be left out. A row that is left out has all four switches off. A name SWAT+ does not know stops the run with `Error: The output object <name> in the input file print.prt is not a valid object`, and a name that appears twice stops it with `print object is duplicated`. Names are case-sensitive.
- `use_obj_lbls = n`: the rows are read in the fixed order of the table below, and the names in the first column are not checked. A missing row or an extra row shifts every later row, so its switches are applied to the wrong output group, with no message. Rows after the end of the file are off. The carbon and gwflow rows cannot be set in this mode.

Order	Row	Output
1	`basin_wb`	basin water balance
2	`basin_nb`	basin nutrient balance
3	`basin_ls`	basin losses from the land
4	`basin_pw`	basin plant and weather
5	`basin_aqu`	basin aquifers
6	`basin_res`	basin reservoirs
7	`basin_cha`	basin legacy channels
8	`basin_sd_cha`	basin SWAT+ channels, channel morphology and channel budget
9	`basin_psc`	basin point sources (recall objects summed)
10 to 17	`region_wb`, `region_nb`, `region_ls`, `region_pw`, `region_aqu`, `region_res`, `region_sd_cha`, `region_psc`	read, but no region files are written in 62.0.1
18	`water_allo`	water allocation, water use and treatment
19 to 22	`lsunit_wb`, `lsunit_nb`, `lsunit_ls`, `lsunit_pw`	landscape unit water balance, nutrient balance, losses, plant and weather
23 to 26	`hru_wb`, `hru_nb`, `hru_ls`, `hru_pw`	HRU water balance, nutrient balance (with the nitrogen cycle file), losses, plant and weather
27 to 30	`hru-lte_wb`, `hru-lte_nb`, `hru-lte_ls`, `hru-lte_pw`	HRU-lte water balance, losses, plant and weather; `hru-lte_nb` is read but writes nothing
31	`channel`	legacy channels
32	`channel_sd`	SWAT+ channels, channel morphology, channel budget, and the sediment budget files per channel, per stream order and for the basin
33	`aquifer`	aquifers
34	`reservoir`	reservoirs and wetlands
35	`recall`	recall objects (point sources and inlets)
36	`hyd`	inflow and outflow hydrographs of every object, and deposition
37	`ru`	routing unit hydrographs
38	`pest`	pesticides in HRUs, channels, reservoirs, aquifers and the basin
39 to 45	`basin_salt`, `hru_salt`, `ru_salt`, `aqu_salt`, `channel_salt`, `res_salt`, `wetland_salt`	salts
46 to 52	`basin_cs`, `hru_cs`, `ru_cs`, `aqu_cs`, `channel_cs`, `res_cs`, `wetland_cs`	constituents; `res_cs` also opens the wetland constituent files, and `wetland_cs` writes nothing

These rows are recognised only when `use_obj_lbls = y`:

Row	Output
`hru_cb_gl`, `hru_cb_trf`, `hru_cb_lyr`, `hru_cb_cpool`, `hru_cb_npool`, `hru_cb_plt`, `hru_cb_flux`, `hru_cb_drv`, `hru_cb_dyn`, `hru_cb_snap`	HRU carbon output, one row per file family. See Carbon outputs.
`lsu_cb_gl`, `lsu_cb_trf`, `lsu_cb_plt`	landscape unit carbon output, area-weighted from the HRUs. See Carbon outputs.
`hru_cb`, `hru_cb_vars`	legacy CSU carbon files, switched with `y` or `1`. See Legacy CSU carbon outputs.
`gwflow_wb`, `gwflow_flux`, `gwflow_heat`, `gwflow_solute`, `gwflow_obs`, `gwflow_pump`	gwflow output. A row that is present replaces the setting from the gwflow input files; a row that is absent leaves it. See gwflow output switches.

Some rows write data only when the project has the matching objects. The `channel_sd` and `basin_sd_cha` files need SWAT+ channels, and the `channel` and `basin_cha` files need legacy channels. The basin landscape files (`basin_wb`, `basin_nb`, `basin_ls`, `basin_pw`) are summed from the landscape unit element file `ls_unit.ele`, and the landscape unit files need `ls_unit.def`; both are named on the `regions` line of `file.cio`.

Print timing

Every output file leaves out the first `nyskip` years of the simulation. After those years:

- Daily rows are written from `day_start` of `yrc_start` until `day_end` of `yrc_end`, every `interval` days, starting with the first day of the window. Output starts on the first simulated day whose day of the year is at least `day_start` in a year at or after `yrc_start`. If the simulation starts in a later year than `yrc_start`, daily output therefore starts on day `day_start` of the first simulated year. Days before it in that year are not written.
- Monthly rows are written at the end of each month, yearly rows at the end of each year, and average annual rows once, at the end of the simulation.

The print window applies to daily rows only. Monthly, yearly and average annual files always cover all years after the warm-up.

Average annual values

The average annual files give the sum over the printed period divided by the number of printed years Y. SWAT+ counts the printed days D (all days after the warm-up) and divides by 365:

$$Y = \frac{D}{365}$$

Leap days are counted in D, so Y is slightly larger than the number of calendar years. When `nyskip` is above 0, 62.0.1 starts the count at the number of simulated years N_{yr} instead of 0:

$$Y = \frac{D + N_{yr}}{365}$$

For a 10-year simulation with 2 warm-up years, Y is about 8.03 instead of 8.00, and the average annual values are about 0.3% lower than the true averages over the printed years. If `nyskip` is equal to or larger than the number of simulated years, no output is written.

Example

`refdata/Ames_sub1/print.prt` in the SWAT+ repository:

```
print.prt:
nyskip      day_start  yrc_start  day_end   yrc_end   interval
0           0               1975         0      2020          1
```

```
aa_int_cnt
0
csvout        use_obj_lbls  cdfout
n             y             n
crop_yld      mgtout        hydcon        fdcout
b             y             n             n
objects                  daily       monthly        yearly         avann
hru_wb                     y             n             n             y
hru_nb                     y             n             n             y
hru_ls                     n             n             n             y
hru_pw                     y             n             n             y
hru_cb                     l             y             y             y
hru_cb_vars                l             y             y             y
```

There is no warm-up period. Daily output covers the whole simulation (1975 to 2020) at an interval of one day. There is no `.csv` output. Crop yields are written both yearly and as average annual values, and the management log `mgt_out.txt` is written.

`use_obj_lbls = y`, so the rows are read by name and every row not listed is off. The HRU water balance, nutrient balance, and plant and weather files are written daily and as average annual values; the HRU losses only as average annual values. The `hru_cb` and `hru_cb_vars` rows switch on the legacy carbon files, with `l` in the daily column for the full set of layer diagnostics. The `hru_cb_vars` files are written only when the carbon model is on (`carbon = 2` in codes.bsn).

`refdata/0su_1hru/print.prt`, written by an older SWAT+ Editor, has `dbout` in place of `use_obj_lbls` with the value `n`, so its rows are read by position. Its list has an `hru_cb` row after `hru_pw`, which is not part of the fixed order: the reader applies that row to `hru-lte_wb` and every later row moves down one group. All rows in that file have the same switches (`n n n y`), so the shift has no visible effect there.

Related files

- file.cio: names this file, the output folder, and the landscape unit files the basin and landscape unit outputs need.
- time.sim: the simulation period that contains the print period.
- Output naming convention: file names and the files each row opens.
- Output reference: contents of each output file.

Sources

- Reader: basin_print_codes_read.f90, called from proc_bsn.f90 after `time_read`
- Type definitions: `type basin_print_codes` and `type print_interval`, and `print_prt_error`, in basin_module.f90
- Print window, warm-up and average annual years: time_control.f90
- Output calls and the `nyskip` test: command.f90
- File opening: output_landscape_init.f90, header_write.f90, header_sd_channel.f90, header_yield.f90, header_mgt.f90, header_hyd.f90
- gwflow overrides: gwflow_read.f90
- Legacy carbon flags: carbon_legacy_module.f90
- Region output (not called): lsreg_output.f90

Column or row	Fortran variable
nyskip, day_start, yrc_start, day_end, yrc_end, interval	pco%nyskip, pco%day_start, pco%yrc_start, pco%day_end, pco%yrc_end, pco%int_day
aa_int_cnt and end years	pco%aa_numint, pco%aa_yrs
csvout, use_obj_lbls, cdfout	pco%csvout, pco%use_obj_labels, pco%cdfout
crop_yld, mgtout, hydcon, fdcout	pco%crop_yld, pco%mgtout, pco%hydcon, pco%fdcout
basin_wb … basin_psc	pco%wb_bsn, nb_bsn, ls_bsn, pw_bsn, aqu_bsn, res_bsn, chan_bsn, sd_chan_bsn, recall_bsn

Column or row	Fortran variable
region_wb … region_psc, water_allo	pco%wb_reg, nb_reg, ls_reg, pw_reg, aqu_reg, res_reg, sd_chan_reg, recall_reg, water_allo
lsunit_*, hru_*, hru-lte_*	pco%{wb,nb,ls,pw}_lsu, pco%{wb,nb,ls,pw}_hru, pco%{wb,nb,ls,pw}_sd
channel, channel_sd, aquifer, reservoir, recall, hyd, ru, pest	pco%chan, sd_chan, aqu, res, recall, hyd, ru, pest
salt rows	pco%salt_basin, salt_hru, salt_ru, salt_aqu, salt_chn, salt_res, salt_wet
constituent rows	pco%cs_basin, cs_hru, cs_ru, cs_aqu, cs_chn, cs_res, cs_wet
hru_cb, hru_cb_vars, hru_cb_<family>, lsu_cb_<family>	pco%cb_hru, pco%cb_vars_hru, pco%cb_<family>_hru, pco%cb_<family>_lsu
gwflow_<part>	pco%gwflow_<part>

Purpose

object.cnt gives the basin name, the basin areas and the number of spatial objects of each type: HRUs, routing units, aquifers, channels, reservoirs, point sources and the others listed below. SWAT+ uses the counts to set up its list of objects before any connection file is read. The file is named in slot 4 of the simulation line of file.cio (default object.cnt). It is required: if it is missing or null, SWAT+ prints Cannot find object.cnt input file and stops.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Line 3 contains 21 values in the order below. The reader does not match columns by header name.

#	Column	Units	Default	Description
1	name	-	blank	Basin name, up to 25 characters. It appears in the name column of the basin output files.
2	ls_area	ha	0	Land area of the basin.
3	tot_area	ha	0	Total area of the basin, including reservoirs and other water bodies.
4	obj	count	0	Total number of spatial objects. Must equal the sum of columns 5 to 18.
5	hru	count	0	Number of HRUs (hru.con).
6	lhru	count	0	Number of HRU-lte objects (hru-lte.con).
7	rtu	count	0	Number of routing units (rout_unit.con).
8	mfl	count	0	Number of gwflow objects (gwflow.con). See gwflow.
9	aqu	count	0	Number of aquifers (aquifer.con).
10	cha	count	0	Number of legacy channels (channel.con).
11	res	count	0	Number of reservoirs (reservoir.con).
12	rec	count	0	Number of recall objects: point sources and inlets (recall.con).
13	exco	count	0	Number of export coefficient objects (exco.con).
14	dlr	count	0	Number of delivery ratio objects (delratio.con).
15	can	count	0	Number of canals. No connection file is read for canals in 62.0.1.
16	pmp	count	0	Number of pumps. No connection file is read for pumps in 62.0.1.
17	out	count	0	Number of outlets (outlet.con).
18	lcha	count	0	Number of channels routed with the SWAT+ channel method (chandeg.con).

#	Column	Units	Default	Description
19	aqu2d	count	0	Number of 2D aquifers. Read and not used.
20	hrd	count	0	Number of herds. Read and not used.
21	wro	count	0	Number of water rights objects. Read and not used.

The column headers are the ones the SWAT+ Editor writes. The old GitBook documentation calls column 8 `gwfl`. Only the order matters to SWAT+.

Object counts

Each count must match the number of objects in the corresponding connection file named on the `connect` line of `file.cio`. SWAT+ numbers all objects in one list, in this order: HRUs, HRU-lte objects, routing units, gwflow objects, aquifers, legacy channels, reservoirs, recall objects, export coefficient objects, delivery ratio objects, canals, pumps, outlets and SWAT+ channels. The object list has `obj` entries, so `obj` must equal the sum of these counts. Counts for canals and pumps reserve places in the list, but no objects are read into them.

If line 3 ends early, the reader reaches the end of the file and the missing counts stay at zero. The `refdata/Osu_1hru` example uses this and leaves out `hrd` and `wro`. If the file has more lines after line 3, SWAT+ keeps reading them in groups of title, header and values, and the last complete group is the one used. A file with a single data line avoids this.

Basin areas

`ls_area` and `tot_area` are not used to compute HRU or object areas; those come from the connection files. They are used in these places:

- `ls_area`: the HRU area when a decision table changes the fraction of the basin an HRU covers, and the HRU areas of the regions in soft calibration (`water_balance.sft`, see soft calibration).
- `tot_area`: the area weights for the basin aquifer output (`basin_aqu_*` files), and the conversion of gwflow basin water balance volumes to depths.

gwflow

When `gwflow = 1` in `codes.bsn`, SWAT+ looks for `chancell.gw` in the run folder after reading `object.cnt`:

- If `chancell.gw` exists and `mfl` is 0, SWAT+ counts the channel cell rows in `chancell.gw` and uses that count as the number of gwflow objects. It adds that count to `obj`, removes the aquifers from `obj`, sets the aquifer count to 0 and reads the gwflow connections from `gwflow.con` in place of `aquifer.con`.
- If `chancell.gw` exists and `mfl` is already above 0, SWAT+ keeps `mfl` and `obj` as given and sets the aquifer count to 0. In that case `obj` should already leave out the aquifers.
- If `chancell.gw` is missing or has no data after its header, SWAT+ turns gwflow off and the run uses the lumped aquifers.

In all three cases SWAT+ opens the `gwflow_record` log file. See gwflow for the gwflow input files.

Example

`refdata/Ames_sub1/object.cnt` in the SWAT+ repository:

```
object.cnt:
name                    ls_area      tot_area       obj       hru      lhru       rtu       mfl       aqu
demo                       1.           1            12         12         0         0         0         0
```

The basin is named `demo` and both areas are 1 ha. The project has 12 objects, all of them HRUs, so `obj` equals `hru`. There are no routing units, aquifers or channels.

`refdata/Osu_1hru/object.cnt` has 4 objects: one HRU, one routing unit, one aquifer and one SWAT+ channel (`lcha`). Its data line stops after `aqu2d`, and the `hrd` and `wro` counts are taken as zero.

Related files

- `file.cio`: names this file and the connection files.
- Connection files: one row per object; the number of rows must match the counts here.
- `codes.bsn`: its `gwflow` code starts the gwflow check above.
- gwflow: `chancell.gw` and the other gwflow inputs.

Sources

- Reader: `basin_read_objs.f90`, called from `proc_bsn.f90` after `codes.bsn`
- Type definitions: `type basin_inputs` in `basin_module.f90`, `type spatial_objects` in `hydrograph_module.f90`
- Object numbering and connection files: `hyd_connect.f90`
- Uses of the areas: `hru_fr_change.f90`, `lcu_read_softcal.f90`, `basin_aquifer_output.f90`, `gwflow_output.f90`
- File name slot: `in_sim%object_cnt` in `input_file_module.f90`

Column	Fortran variable
name	bsn%name
ls_area	bsn%area_ls_ha
tot_area	bsn%area_tot_ha
obj	sp_ob%objs
hru	sp_ob%hru
lhru	sp_ob%hru_lte
rtu	sp_ob%ru
mfl	sp_ob%gwflow
aqu	sp_ob%aqu
cha	sp_ob%chan
res	sp_ob%res
rec	sp_ob%recall
exco	sp_ob%exco
dlr	sp_ob%dr
can	sp_ob%canal
pmp	sp_ob%pump
out	sp_ob%outlet
lcha	sp_ob%chandeg
aqu2d	sp_ob%aqu2d
hrd	sp_ob%herd
wro	sp_ob%wro